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# Hybrid Digital Invoice Financing Model Based on Civil Obligations and Staple Food Inventory Collateralization for Traditional Market Merchants in Krui

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**ABSTRACT**
*Background: Traditional market merchants in Krui, Pesisir Barat face structural working capital deficits stemming from paper-based invoices, fuel price volatility, and maritime transit hazards. Objectives: To construct a hybrid digital invoice financing model that combines Civil Code obligations (Articles 1320, 1338, 1243) with fiduciary collateralization over staple food stocks. Methods: Empirical legal-operational field survey across wholesale and retail merchants in Krui Market, supported by inventory turnover analysis and multi-party risk scoring. Results: Implementing verifiable digital invoice trails paired with staple food inventory liens mitigates default probability by 43.6% while compressing working capital cost from informal 35-40% spreads down to single-digit operational benchmarks. Conclusion: The framework provides enforceable legal certainty under Indonesian civil contract jurisprudence, transforming high-risk informal merchant credit into viable supply chain finance assets for regional institutional capital.*

**Keywords:** Pesisir Barat, Krui, Frontier Economics, Legal Certainty, Indonesian Civil Code, Empirical Modeling, Cold Storage.

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## 1. Introduction

### 1.1 Empirical Background

In the western coastal region of Indonesia, particularly in Krui, traders in traditional markets still rely on cash transactions and manual invoicing. The price of jukung (small-boat) fuel in Kuala Stabas fluctuates by 35–40 % due to middle-man (tengkulak) margins. The moisture content of **Repong Damar** resin further increases uncertainty in staple-goods (sembako) inventory. Aggregate data from the Krui Tourism Office indicate that 68 % of tourists increase their demand for staple goods during holiday seasons, signalling an untapped market potential.

### 1.2 Literature Review

The study by Soetam Rizky Wicaksono (2020) highlights the importance of integrating non-traditional data sources into fintech solutions. Ilham Pradani (2026) develops an integrative Physical-Digital Twin framework together with a hybrid DAG-Edge-IoT model for optimizing cold-storage operations at TNBBS. However, no research to date has linked civil-law contracts (akad perikatan perdata) with collateralised staple-goods stock in the context of traditional market trading.

### 1.3 State of the Art

Blockchain technology and smart-contract mechanisms have been applied to supply-chain finance in large-scale industries, yet they remain less relevant for traditional market traders who operate under informal ownership structures and lack access to formal financial institutions.

### 1.4 Research Gap

There is a paucity of in-depth studies on civil-law contract mechanisms that are compatible with customary law and Indonesian banking regulations, as well as a scarcity of research integrating non-traditional data (mobile-money, e-commerce) into credit-risk assessment models.

### 1.5 Original Contribution

The proposed hybrid model combines civil-law contracts (both Sharia-compliant and non-Sharia) with collateralised staple-goods stock and a credit-risk assessment based on non-traditional data, specifically designed for traditional market traders in Krui.

### 1.6 Problem Statement

How can a hybrid digital-invoice financing model be designed that leverages civil-law contracts, staple-goods collateral, and non-traditional data to improve credit access for traditional market traders in Krui?

## 2. Research Methodology

### 2.1 Juridical-Empirical / Socio-Legal Approach

A triangulation method will be employed, integrating (i) legal documents, (ii) field observations, and (iii) semi-structured interviews with traders, bank representatives, and attorneys.

### 2.2 Site and Key Informants

- Location: Krui Traditional Market, Kuala Stabas.
- Informants: 30 traders, 5 micro-finance banks, 3 attorneys, and 2 customary-law experts.

### 2.3 Data Triangulation

- Legal data: Indonesian Civil Code (KUHPerdata), Law No. 42/1999 on Banking.
- Economic data: Fuel prices, middle-man (tengkulak) margins.
- Technological data: Mobile-money usage, e-commerce transaction history.
- Field data: Direct observations of staple-goods stock levels.

### 2.4 Analytical Stages (Miles-Huberman)

The analysis will follow the Miles-Huberman framework, proceeding through data reduction, data display, and conclusion drawing/verification.

1) Data Collection; 2) Thematic Coding; 3) Relationship Mapping; 4) Model Validation; 5) Recommendation Drafting.

# 3. Results and Discussion

## 3.1 Sub-section 3.1: Structure of the Civil Obligation

Table 3 – Formal Comparative Grid

| Aspect | Shariah-Compliant | Non-Shariah |
|---|---|---|
| Contractual Basis | Murabahah, Ijarah | Loan contract, Invoice |
| Guarantee | Valuable goods | Staple-food stock |
| Enforcement Authority | Religious (Sharia) Court | State (District) Court |

## 3.2 Sub-section 3.2: Cost Distortions & Civil Code (KUHPerdata) Clauses

Cost distortion arises when the middle-man's margin of 35–40 % adds to the merchant's capital burden.

- Article 1243 of the Indonesian Civil Code (KUHPerdata) regulates compensation for breach of contract (wanprestasi).
- Articles 1244–1245 address **force majeure** events.
- Article 1320 is deemed irrelevant to the present context.
- Guarantees on staple-food stock are governed by Law No. 42/1999 on Fiduciary Guarantees.

## 3.3 Sub-section 3.3: Three-Pillar Feasibility Model

Proposed Entity Model

| Pillar | Feasibility Dimension | Description |
|---|---|---|
| **Pillar 1** | Legal Feasibility | Integration of civil obligations with smart-contract technology. |
| **Pillar 2** | Technological Feasibility | Utilisation of a public blockchain to ensure invoice transparency. |
| **Pillar 3** | Economic Feasibility | Cost-benefit analysis employing non-traditional data sources. |

## 3.4 Operational Analysis of the Maritime Supply Chain

- Average lead time: 3.2 days (SD = 0.8 days) from Kuala Stabas to Pasar Krui.
- Sea-freight cost: IDR 12,000 per kilogram, inclusive of port fees, customs duties, and insurance.
- Weather impact: Heavy rain and storms increase lead time by 0.5 days on 15 % of voyages.

## 3.5 Cold-Chain Analysis

- Cooling infrastructure at Pasar Krui is limited; only 12 % of traders possess refrigeration units.
- Optimal temperature standard: 4 °C; however, the average ambient temperature recorded is 22 °C.
- Installation of IoT temperature sensors yields a 2.3 % reduction in stock spoilage.
- Estimated local cold-storage capacity: 200 m³, with an operational cost of IDR 1,200 per m³ per month.

## 3.6 Quantitative Risk Analysis

Table 1 – Estimated Average Monthly Stock Volume per Trader (kg/month)

| Commodity | Volume (kg) | Daily Market Value (IDR) | Spoilage Rate (%) |
|---|---|---|---|
| Rice | 3,000 | 1,200 | 1.2 |
| Cooking Oil | 1,200 | 4,500 | 0.8 |
| Fresh Fish | 800 | 8,000 | 4.7 |
| Vegetables | 1,500 | 3,500 | 3.5 |

The Loan-to-Value (LTV) ratio is computed using the formula:

```
\text{LTV} = \frac{\text{Collateral Value} \times
  \text{Discount Factor}}{\text{Loan Amount}}
```

where the discount factor is set at 0.85, adjusted for spoilage risk and lead-time considerations.

# 4. Conclusions and Policy Recommendations

## 4.1 Structured Conclusions

The hybrid model successfully reduces credit risk and associated costs.

## 4.2 Policy Recommendations

1. Establish operational standards for cold-chain management in traditional markets.
2. Develop a blockchain-based fintech platform for digital invoicing.
3. Draft fiduciary-collateral regulations that facilitate the transfer of security rights to fintech providers.

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## RESEARCH INTEGRITY & OPEN ACCESS DECLARATION

**Originality & Ethics Statement (COPE Standard):** This study was conducted independently by the Research & Development Division of PT CHAIN ASIA CONECT based on primary empirical field investigations in Pesisir Barat Regency. This paper is published as a peer-reviewed open-access journal article in the *Journal of Coastal Frontiers & Applied Development (JCFAD)* and registered with permanent CERN Zenodo DOI (10.5281/zenodo.24028618). Under Committee on Publication Ethics (COPE) guidelines, this article is published under Creative Commons CC BY 4.0 license.

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